```
fs =["apple", "banana",
     "orange"]
ndex, fruit in enumerate(fruit
print(index, fruit)
```

```
def multiply(a, b):
    return(a*b)
answer = multiply(3, 4)
print(answer)
```

WK3

```
def show(*args, **kwargs):
    print(args)
    print(kwargs)
show(4, 7, colour="blue", size=
     "large")
```

objects. __init__ b) runs
self refers to the
an. self.name = name
t. Attributes are
Each object has its
ng one objects attribute
another. A method
class. Call it using
od can call another
an update an objects
attributes shared by
object has its own
e. You can store
gh them to call a

```
ass Cat:
    def speak(self):
        print("Meow")

Cat = Cat()
Cat.speak() -> Meow

s = [Animal(), Animal()
]
mal in animals:
nimal.speak()
```

WK5

Series and dataframe are used to store and organise data. pd.Series() creates a single column of data. pd.Dataframe creates tables with rows and columns. dtypes shows the type of data. shape shows the number of rows and columns. pd.read_csv() reads a csv file df["column"] selects one column df[["a","b"]] selects multiple columns. iloc[] selects using row/column positions. loc[] selects using labels. df[df["Age"] >= 100] filters rows. | means OR & means AND. ascending=True means smallest to largest. ascending=false means largest to smallest. We use .agg() to use multiple calculations at ~~once~~ once and groupby() to organise rows into groups. value_counts() how many times a value appears. apply() applies a function to a value in a column. lambda creates a small function without giving it a name. if and else allow conditions. describe() gives statistical information. to_csv() save a dataframe to a csv file. plot creates a graph.

```
import pandas as pd
s = pd.Series([5, 10, 15])
df = pd.Dataframe({"Name": ["Amy", "Ben"],
                   "Age": [18, 21]})
df.dtypes   df.iloc[0, 1]
df.shape    df.loc[0, "Name"]
df["total"] = df["price"] + df["tax"]
df.sort_values("Age", ascending=Tru
df.groupby("city")["Salary"].agg(["mean"])
df["Age"].plot(kind="bar")
df[df["Ag"]>=19]
df["city"].value_
    counts()
df.describe()
```

WK6